JAMA Network Open

Original Investigation | Surgery

Smoking Behavior and Major Cardiovascular and Limb Events in Patients With Peripheral Artery Disease

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Abstract

IMPORTANCE Smoking is the strongest modifiable risk factor for peripheral arterial disease (PAD), yet evidence on how postdiagnosis smoking behavior changes affect clinical outcomes remains limited.

OBJECTIVE To evaluate the association among smoking cessation, reduction, and continuation after PAD diagnosis and subsequent cardiovascular and limb-specific outcomes.

DESIGN, SETTING, AND PARTICIPANTS This nationwide, population-based, retrospective cohort study assessed adults 40 years or older newly diagnosed with PAD from January 1, 2011, to December 31, 2018, with follow-up through 2021, who reported active smoking before diagnosis and completed health examinations before and after diagnosis. Data were from the Korean National Health Insurance Service–National Health Information Database. Data analysis was conducted from December 2023 to November 2025.

EXPOSURES Postdiagnosis smoking behavior, classified as quitters (complete cessation), reducers (≥20% decrease in daily cigarette consumption), or maintainers (<20% decrease or any increase in consumption [reference group]).

MAIN OUTCOMES AND MEASURES Main outcomes were major adverse cardiovascular and cerebrovascular events (MACCEs), including all-cause death, myocardial infarction, coronary revascularization, and ischemic stroke, and major adverse limb events (MALEs), including limb revascularization or major amputation. Inverse probability of treatment-weighted hazard ratios (HRs) were estimated using Cox proportional hazards regression models for MACCEs and Fine-Gray competing risk models for MALEs.

RESULTS Among 189 545 active smokers with incident PAD (mean [SD] age, 54.2 [11.8] years; 167 659 [88.5%] male), 38 952 (20.6%) quit smoking, 38 709 (20.4%) reduced consumption, and 111 884 (59.0%) maintained smoking. Compared with maintainers, quitters had significantly lower risks of MACCEs (17.8 vs 20.2 per 1000 person-years; HR, 0.88; 95% CI, 0.84-0.91) and MALEs (1.4 vs 1.8 per 1000 person-years; HR, 0.83; 95% CI, 0.73-0.94). Reducers showed no significant difference in MACCEs (HR, 1.00; 95% CI, 0.96-1.04) or MALEs (HR, 0.92; 95% CI, 0.81-1.04). Dose-response analysis showed risk reduction only with near-complete cessation. Findings were consistent across subgroups, with the largest reductions observed for myocardial infarction.

CONCLUSIONS AND RELEVANCE In this cohort study of 189 545 active smokers with newly diagnosed PAD, complete smoking cessation was associated with lower risks of both cardiovascular

(continued)

Key Points

Question Is partial smoking reduction after peripheral arterial disease (PAD) diagnosis associated with improved outcomes or is complete cessation required?

Findings In this nationwide cohort study of 189 545 active smokers with newly diagnosed PAD, complete smoking cessation was associated with a 12% lower risk of cardiovascular events and 17% lower risk of limb events compared with continued smoking. Partial smoking reduction provided no significant benefit for either outcome.

Meaning Complete smoking cessation, rather than reduction, should be emphasized as the primary therapeutic goal in secondary prevention for patients with PAD.

+ Invited Commentary

+ Supplemental content

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Abstract (continued)

and limb-specific adverse events, whereas partial smoking reduction showed no significant benefit. These findings support complete cessation as the primary therapeutic target in patients with PAD.

JAMA Network Open. 2026;9(8):e2629337. doi:10.1001/jamanetworkopen.2026.29337

Introduction

Peripheral arterial disease (PAD) represents a significant global health burden, with the total number of patients with PAD nearly doubling during the past 3 decades[1] and projected to reach 360 million cases by 2050 as increasing metabolic risk factors compound demographic shifts.[2] PAD not only diminishes quality of life through claudication, ischemic ulcers, and rest pain[3-6] but also remains a major cause of nontraumatic lower-extremity amputation.[7] Furthermore, patients with PAD face a 2- to 3-fold increased risk of major cardiovascular events and related mortality,[8,9] underscoring the systemic nature of atherosclerotic disease.

Among modifiable risk factors, tobacco use remains the strongest risk factor associated with PAD development, progression, and adverse outcomes,[10-12] with a stronger association compared with that for other atherosclerotic diseases.[13] Accordingly, smoking cessation is universally recommended as a primary intervention for patients with PAD across all disease stages, receiving a Class I recommendation in multiple clinical practice guidelines.[14-17] Although prior studies[10-12] consistently demonstrate that baseline smoking status—typically assessed cross-sectionally at diagnosis or intervention—is associated with disease severity and outcomes, data evaluating the impact of longitudinal smoking behavior changes after diagnosis are limited. Available data are largely derived from single-center, observational studies with modest sample sizes and limited follow-up,[18-20] precluding granular analysis of the association between the extent of smoking reduction and various clinical outcomes. Therefore, we conducted a nationwide, population-based cohort study to examine the association of postdiagnosis smoking cessation and reduction on cardiovascular and limb-specific outcomes in patients with PAD. We further examined whether these associations persist across demographic and comorbidity subgroups.

Methods

Study Design

We conducted a retrospective cohort study of patients newly diagnosed with PAD using the Korean National Health Insurance Service–National Health Information Database (NHIS-NHID). Active smokers prior to the diagnosis of PAD were categorized according to postdiagnosis smoking behavior changes based on the self-reported survey results from health checkups, and between-group differences in cardiovascular and limb-related outcomes were analyzed (eFigure 1 in Supplement 1). Our primary interest was to examine whether the risk of adverse outcomes differs among those who quit smoking (quitters) or reduced smoking (reducers) compared with those who maintained their smoking habits (maintainers) after PAD diagnosis as a reference. This study was approved by the institutional review board of Seoul National University Hospital and conducted in accordance with the Declaration of Helsinki.[21] The requirement for written informed consent was waived by the board because the study used deidentified claims data from NHIS-NHID. The study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cohort studies.

Data Source

The Korean NHIS is a nationwide insurance system covering approximately 97% of the South Korean population.[22] The NHIS-NHID provides linked data on health care utilization (diagnostic codes,

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prescriptions, procedures, and mortality) and biennial national health screening programs (participation rate, 76.9% in 2018), which include self-reported health behaviors, anthropometric measurements, and laboratory tests.[23,24] Detailed data source information is provided in eMethods 1 in Supplement 1.

Study Population

Our population of interest included patients diagnosed with PAD from January 1, 2011, to December 31, 2018. Patients were identified using *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10)* codes I70.2 (atherosclerosis of native arteries of the extremities) or I73.9 (peripheral vascular disease, unspecified), recorded as the primary diagnosis or among up to 4 secondary diagnoses, with at least 1 concurrent antithrombotic prescription on the same claim date. The *ICD-10* schema was adapted from previous PAD research[25,26] but restricted a priori by excluding code I73.8, which captures nonatherosclerotic vasospastic or microvascular conditions, and codes I74.3 to I74.5, which may include acute embolic or thrombotic arterial occlusions distinct from chronic atherosclerotic PAD. Antithrombotic medications included antiplatelet agents, oral anticoagulants, and vasoactive agents used in PAD management (eTable 1 in Supplement 1). Those who had undergone at least one health examination before PAD diagnosis and a subsequent examination afterwards were eligible for the study. The presence of smoking data on both examinations was required for inclusion. To quantify the association of smoking cessation behavior, only active smokers at the prediagnosis health examination were included in the primary analysis; therefore, individuals who were nonsmokers before diagnosis, including never-smokers and former smokers, were not eligible. To avoid reverse causality, we also excluded patients with (1) a previous PAD diagnosis between 2002 and 2010, (2) missing or erroneous smoking data, and (3) a primary outcome occurring between the prediagnosis and postdiagnosis health examinations. The patient selection process is illustrated in eFigure 2 in Supplement 1.

Smoking Behavior

Smoking status was determined using self-reported survey data from the structured NHIS health examination questionnaires. The study cohort of current smokers at the prediagnosis examination was classified into 3 groups according to postdiagnosis smoking behavior: (1) quitters (reported smoking cessation), (2) reducers (continued smoking with a ≥20% reduction in daily cigarette consumption), and (3) maintainers (continued smoking with a <20% reduction, no change, or any increase in daily cigarette consumption).

Outcome Measures

The primary outcome was major adverse cardiovascular and cerebrovascular events (MACCEs), a composite of all-cause death, myocardial infarction, coronary revascularization, and ischemic stroke. Secondary outcomes included major adverse limb events (MALEs), defined as a composite of lower-extremity revascularization or major amputation. Outcome events were adjudicated using *ICD-10* codes in combination with procedural or imaging codes and, where applicable, inpatient admission status, as detailed in eTable 1 in Supplement 1. For outcome evaluation, patients were followed up from the date of postdiagnosis health examination until the occurrence of an outcome event, death, or the end of the study period (December 31, 2021), whichever came first.

Statistical Analysis

Detailed statistical methods are provided in eMethods 2 in Supplement 1, with key elements summarized below. Baseline characteristics were summarized using descriptive statistics. Incidence rates for primary and secondary outcomes were calculated per 1000 person-years.

To minimize selection bias and balance covariates across smoking behavior groups, we used inverse probability of treatment weighting (IPTW), with propensity scores estimated using generalized boosted models. Covariates included demographic factors, comorbidities, medication

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history, anthropometric and laboratory parameters, and health behaviors (detailed in eMethods 2 and eTable 2 in Supplement 1). For MACCEs, hazard ratios (HRs) and 95% CIs were estimated using Cox proportional hazards regression. For MALEs, HRs were estimated using the Fine-Gray subdistribution hazards model, treating death as a competing event. The proportional hazards assumption was assessed using time-covariate interaction terms, and no statistically significant interactions were detected. Doubly robust estimation combining IPTW and adjustment for all baseline covariates (demographics, comorbidities, medication history, anthropometric and laboratory parameters, and health behaviors) was performed to address potential residual confounding, yielding adjusted hazard ratios (AHRs). Dose-response relationships were examined using restricted cubic splines.[27] Cumulative incidence curves were constructed in the IPTW-weighted cohort: weighted Kaplan-Meier methods for MACCEs and cumulative incidence functions under a competing risk framework, with death as a competing event, for MALEs.

The primary analyses were conducted using complete case analysis. To assess the robustness of our findings under different assumptions about missing data, we performed sensitivity analyses using imputation and missing indicator methods. We also performed sensitivity analyses using alternative thresholds for the reducer category (≥25% and ≥50% reductions in daily cigarette consumption) and excluding patients with any history of myocardial infarction, ischemic stroke, or coronary revascularization before PAD diagnosis. Predefined subgroup analyses with interaction testing were performed to evaluate effect modification across clinical strata.

Statistical tests were 2-sided, with significance set at $P < .05$. Analyses were performed from December 2023 to November 2025 using SAS, version 9.4 (SAS Institute Inc) and R, version 4.1.0 (R Foundation for Statistical Computing).

Results

Study Population

Of 196 882 eligible patients with incident PAD who reported active smoking at prediagnosis health examinations conducted within 2 years before diagnosis, 189 545 (96.3%; mean [SD] age, 54.2 [11.8] years; 167 659 [88.5%] male and 21 886 [11.5%]) had complete covariate data and comprised the primary analysis cohort (**Table 1**; eFigure 2 in Supplement 1). Based on subsequent health examination results, patients were classified into 3 groups according to their smoking behavior change after PAD diagnosis: 38 952 quitters (20.6%), 38 709 reducers (20.4%), and 111 884 maintainers (59.0%).

Compared with maintainers and reducers, quitters were older, more likely to be female, and had a higher Charlson comorbidity index (Table 1). Quitters also had fewer smoking pack-years, lower prediagnosis daily cigarette consumption, and lower rates of alcohol consumption. No differences in baseline socioeconomic status, residential area, physical activity, body mass index, waist circumference, fasting glucose status, or blood pressure were observed (complete list presented in eTable 3 in Supplement 1). To adjust for between-group differences, we applied IPTW, which achieved balance across all covariates (standardized mean difference, <0.1) (eTable 4 in Supplement 1). The median (IQR) interval from the prediagnosis health examination to PAD diagnosis was 292 (149-463) days and from PAD diagnosis to the postdiagnosis examination was 304 (161-477) days, with comparable distributions across smoking behavior groups (eTable 5 in Supplement 1).

Cardiovascular and Limb Outcomes

During a mean (SD) follow-up of 5.0 (2.2) years for MACCEs and 5.1 (2.2) years for MALEs, a total of 18 502 MACCE and 1596 MALE events occurred. Overall, complete cessation, but not smoking reduction, was associated with lower cardiovascular and limb event risk compared with maintenance of smoking (**Table 2**). For MACCEs, quitters had a lower incidence than maintainers (17.8 vs 20.2 per 1000 person-years; HR, 0.88; 95% CI, 0.84-0.91), whereas reducers showed no significant difference (20.2 per 1000 person-years; HR, 1.00; 95% CI, 0.96-1.04). Similarly, for MALEs, quitters

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had a significantly lower incidence rate (1.4 vs 1.8 per 1000 person-years; HR, 0.83; 95% CI, 0.73-0.94; *P* = .005) compared with maintainers. Reducers showed no significant difference in risk (1.6 per 1000 person-years; HR, 0.92; 95% CI, 0.81-1.04; *P* = .18). Cumulative incidence curves demonstrated clear separation between quitters and maintainers for both outcomes, whereas reducers tracked closely with maintainers (**Figure 1**).

Dose-Response Analysis

To examine the dose-response relationship between smoking reduction and outcomes, we performed restricted cubic spline analyses (**Figure 2**). In analyses of percentage of smoking reduction, HRs for MACCEs remained close to 1.0 for mild-to-moderate reductions, decreasing gradually with greater reduction and reaching statistical significance at approximately 75%

Table 1. Prediagnosis Characteristics of Active Smoking Patients With Incident Peripheral Artery Disease by Postdiagnosis Smoking Behavior Change

Characteristic	No. (%) of patients[a]			Maximum SMD[b]
	Quitters (n = 38 952)	Reducers (n = 38 709)	Maintainers (n = 111 884)	
Age, mean (SD), y	56.5 (12.1)	54.6 (11.9)	53.2 (11.5)	0.276
Sex				
Female	6735 (17.3)	4334 (11.2)	10 817 (9.7)	0.225
Male	32 217 (82.7)	34 375 (88.8)	101 067 (90.3)	−0.225
Charlson Comorbidity Index, mean (SD)	1.0 (1.4)	0.9 (1.3)	0.8 (1.2)	0.157
Medical history				
Myocardial infarction	133 (0.3)	147 (0.4)	421 (0.4)	−0.006
Stroke	493 (1.3)	443 (1.1)	1206 (1.1)	0.017
Coronary revascularization	377 (1.0)	359 (0.9)	982 (0.9)	0.009
Peripheral revascularization	738 (1.9)	484 (1.3)	801 (0.7)	0.104
Limb amputation	14 (0.04)	8 (0.02)	11 (0.01)	0.017
Comorbidities				
Hypertension	18 901 (48.5)	17 553 (45.4)	48 098 (43.0)	0.111
Diabetes without complications	12 960 (33.3)	12 221 (31.6)	32 897 (29.4)	0.083
Diabetes with complications	7090 (18.2)	6949 (18.0)	18 719 (16.7)	0.039
Heart failure	2689 (6.9)	2147 (5.6)	5178 (4.6)	0.098
End-stage kidney disease	95 (0.2)	56 (0.1)	92 (0.1)	0.040
Comedication				
Statin	14 010 (36.0)	12 998 (33.6)	35 795 (32.0)	0.084
Oral anticoagulants	722 (1.9)	491 (1.3)	973 (0.9)	0.085
Single antiplatelet[b]	22 821 (58.6)	23 047 (59.5)	65 933 (58.9)	−0.019
Dual antiplatelets[b]	1606 (4.1)	1153 (3.0)	2448 (2.2)	0.111
Antithrombotic agents	24 546 (63.0)	24 296 (62.8)	68 575 (61.3)	0.036
Cumulative smoking pack-years				
<10	12 916 (33.2)	6265 (16.2)	26 255 (23.5)	0.458
10 to <20	9617 (24.7)	8766 (22.7)	31 015 (27.7)	
20 to <30	6748 (17.3)	8018 (20.7)	23 921 (21.4)	
≥30	9671 (24.8)	15 660 (40.5)	30 693 (27.4)	
Daily cigarette consumption				
<10	9271 (23.8)	3025 (7.8)	14 541 (13.0)	0.555
10 to <20	15 610 (40.1)	12 816 (33.1)	47 638 (42.6)	
≥20	14 071 (36.1)	22 868 (59.9)	49 705 (44.4)	
Alcohol consumption				
Nondrinker	17 145 (44.0)	13 073 (33.8)	34 258 (30.6)	0.298
Low-risk drinker	18 649 (47.9)	21 373 (55.2)	62 672 (56.0)	
Moderate-risk drinker	2007 (5.2)	2768 (7.2)	9365 (8.4)	
Heavy-risk drinker	1151 (3.0)	1495 (3.9)	5589 (5.0)	

Abbreviation: SMD, standardized mean difference.

[a] Unless otherwise indicated.

[b] Maximum SMD across all pairwise comparisons in the 3 smoking behavior groups; values greater than 0.1 indicate meaningful imbalance. Variables with a maximum SMD less than 0.1 (including socioeconomic status, residential area, physical activity, body mass index, waist circumference, fasting glucose status, and blood pressure) are not shown; the complete list is in eTable 3 in Supplement 1. Characteristics shown are unweighted (before inverse probability of treatment weighting); the after before inverse probability of treatment weighting balance is given in eTable 4 in Supplement 1.

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reduction. For MALEs, a gradual downward trend was observed, although CIs remained wide until near-complete cessation. When we modeled absolute postdiagnosis cigarette consumption, the risks of both MACCEs and MALEs increased monotonically with increasing daily consumption, with elevated risks evident even at low levels of continued smoking.

Individual Outcome Components

When we analyzed the individual outcome components of MACCEs and MALEs, complete smoking cessation was consistently associated with lower risk across most cardiovascular and limb events (eFigure 3 and eFigure 4 in Supplement 1). For MACCEs, quitters had a significantly lower risk of death, myocardial infarction, ischemic stroke, and coronary revascularization than maintainers, with HRs ranging from 0.68 (95% CI, 0.60-0.77) to 0.89 (95% CI, 0.84-0.93) (eFigure 3 in Supplement 1). For MALEs, the lower risk was associated most strongly with peripheral revascularization (HR, 0.80; 95% CI, 0.69-0.92). Reducers showed no significant difference except for coronary revascularization. For major amputation, quitters did not have a statistically significant reduction (7 events), whereas reducers had an increased risk (20 vs 31 events) (HR, 1.93; 95% CI, 1.08-3.47).

Sensitivity Analyses

The primary findings of lower risk in quitters, but not in reducers, for both MACCEs and MALEs were similar when using doubly robust estimation, confirming adequate covariate balance achieved through IPTW (Table 2). The findings also remained consistent across sensitivity analyses accounting for missing data (eTable 6 in Supplement 1). Results from multiple imputation and missing indicator methods using doubly robust estimation yielded similar effect estimates to those obtained using complete case analysis for both MACCEs and MALEs. Sensitivity analyses using stricter reducer thresholds (≥25% and ≥50% reductions in daily cigarette consumption) showed no significant reduction in MACCE or MALE risk among reducers compared with maintainers at either threshold (eTable 7 in Supplement 1). In an additional sensitivity analysis excluding patients with prior myocardial infarction, ischemic stroke, or coronary revascularization before PAD diagnosis, results were also consistent with the primary findings: quitters had a lower risk of MACCEs (AHR, 0.86; 95% CI, 0.82-0.89), whereas reducers did not (AHR, 0.99; 95% CI, 0.95-1.03) (eTable 8 in Supplement 1).

Subgroup Analyses

The association between smoking cessation and lower MACCE risk was consistent across most examined subgroups (**Figure 3**), with significant risk reductions observed across sex, age, and comorbidity status. Importantly, this pattern persisted among high-risk patients, including those 65 years or older and those with prior peripheral revascularization, low physical activity, or severe kidney

Table 2. Association Between Postdiagnosis Smoking Behavior Change and Major Adverse Cardiovascular and Limb Events: IPTW and Doubly Robust Analysis

Smoking status	No. of events/total No. of patients (%)[a]	Follow-up duration, mean (SD), y[a]	Incidence per 1000 person-years (95% CI)[a]	IPTW adjusted[a,b]		Doubly robust[a,c]	
				HR (95% CI)	*P* value	AHR (95% CI)	*P* value
MACCEs							
Quitter	3580/38 388 (9.3)	5.23 (2.17)	17.84 (17.26-18.43)	0.88 (0.84-0.91)	<.001	0.86 (0.83-0.89)	<.001
Reducer	3801/38 343 (9.9)	4.92 (2.21)	20.16 (19.53-20.82)	1.00 (0.96-1.04)	.99	0.99 (0.95-1.03)	.58
Maintainer	11 121/111 394 (10.0)	4.94 (2.23)	20.19 (19.82-20.57)	1.00 [Reference]	NA	1.00 [Reference]	NA
MALEs							
Quitter	294/38 388 (0.8)	5.33 (2.14)	1.44 (1.28-1.61)	0.83 (0.73-0.94)	.005	0.80 (0.70-0.92)	.002
Reducer	312/38 343 (0.8)	5.03 (2.19)	1.62 (1.45-1.81)	0.92 (0.81-1.04)	.18	0.90 (0.79-1.02)	.11
Maintainer	990/111 394 (0.9)	5.06 (2.21)	1.75 (1.65-1.87)	1.00 [Reference]	NA	1.00 [Reference]	NA

Abbreviations: AHR, adjusted hazard ratio; HR, hazard ratio; IPTW, inverse probability of treatment weighting; MACCEs, major adverse cardiovascular and cerebrovascular events; MALEs, major adverse limb events; NA, not applicable.

[a] Data are from the complete case cohort (n = 189 545) with IPTW applied to balance baseline covariates.

[b] HRs derived from IPTW-weighted Cox proportional hazards regression models.

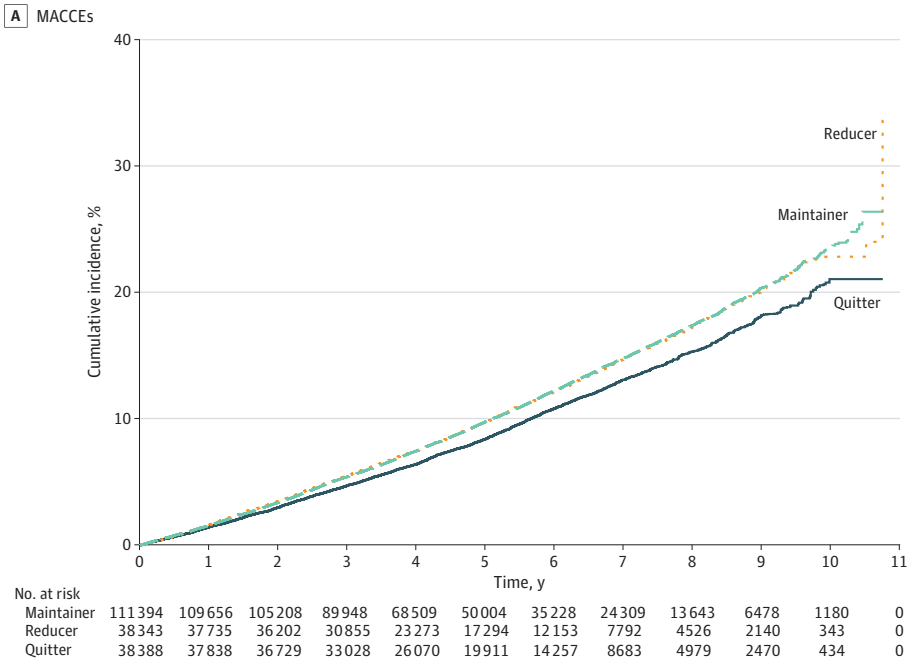
[c] Doubly robust HRs derived from weighted Cox proportional hazards regression models with additional covariate adjustment.

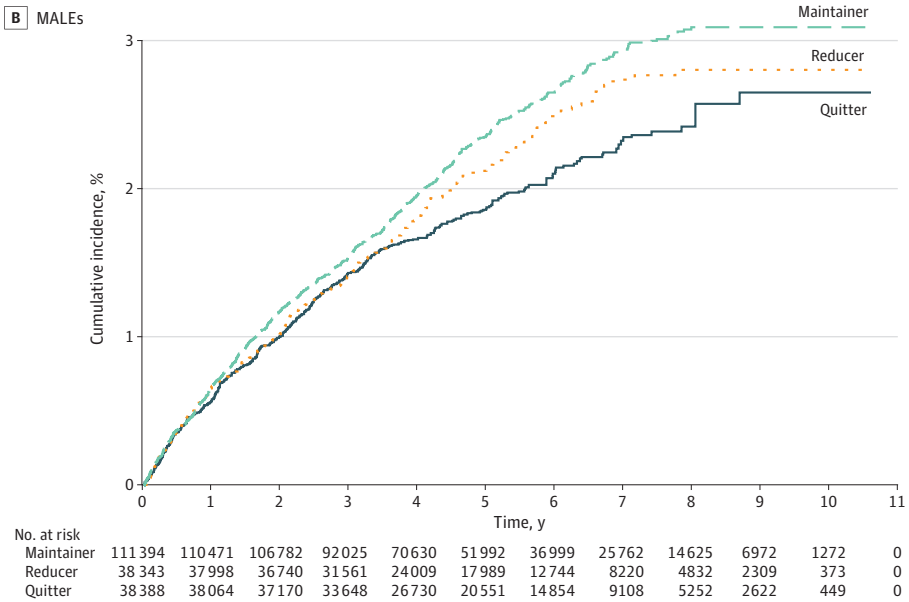
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impairment (event rates, 12%-44%). Among these high-risk groups, smoking cessation showed a consistent association with lower risk, with HRs ranging from 0.66 to 0.90, although the association was less precise in the smaller severe kidney impairment subgroup (HR, 0.77; 95% CI, 0.57-1.03). Conversely, smoking reduction was not associated with significant risk reduction in any subgroup examined.

We further examined whether the association of smoking behavior change with MACCEs differed across patient subgroups using interaction terms (Figure 3). Two significant effect modifications were identified. First, the association between smoking cessation and MACCEs varied

Figure 1. Kaplan-Meier Curves Showing Cumulative Incidence of Major Adverse Cardiovascular and Limb Events by Smoking Behavior Change in Patients With Peripheral Artery Disease





Among active smokers with peripheral artery disease, cumulative incidence curves are shown for major adverse cardiovascular and cerebrovascular events (MACCEs) and major adverse limb events (MALEs) according to postdiagnosis smoking behavior change. Numbers at risk are shown below each panel. Log-rank $P < .001$ for both comparisons.

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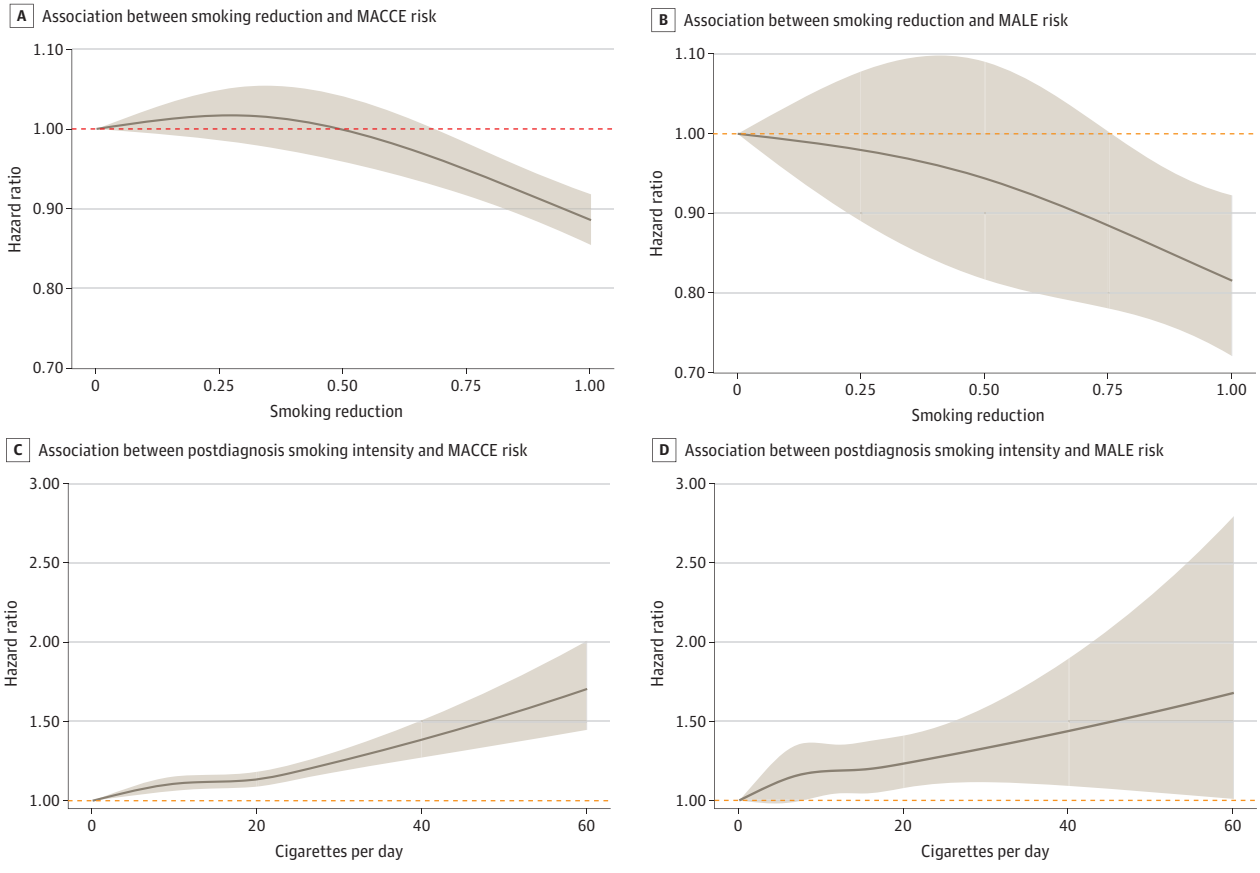
by physical activity level (*P* for interaction = 0.04); cessation showed greater relative risk reduction among physically active patients (HR range, 0.78 [95% CI, 0.69-0.87] to 0.82 [95% CI, 0.75-0.90] across activity subgroups) compared with patients with a sedentary lifestyle (HR, 0.90; 95% CI, 0.86-0.95) (Figure 3). Second, among patients with prior peripheral revascularization, cessation was associated with a markedly larger risk reduction (HR, 0.66; 95% CI, 0.54-0.81); notably, smoking reduction reached statistical significance only in this subgroup (HR, 0.79; 95% CI, 0.63-0.99) (*P* for interaction = 0.04).

For MALEs, quitting smoking also demonstrated a consistent pattern of overall benefit across subgroups, although with wider CIs reflecting the lower overall event rate (eFigure 5 in Supplement 1). Smoking reduction showed no consistent association with lower risk. A significant interaction was observed for prior MACCE history; however, the elevated hazard among reducers in this subgroup (HR, 1.79; 95% CI, 1.14-2.82) should be interpreted cautiously, given the limited number of events (n = 31).

Discussion

In this nationwide cohort study leveraging serial health examinations from approximately 190 000 active smokers with newly diagnosed PAD, complete smoking cessation after diagnosis was associated with a 12% relative reduction in MACCE and a 17% reduction in MALE risks compared with those associated with continued smoking. In contrast, smoking reduction without complete

Figure 2. Dose-Response Relationship Between Smoking Behavior and Risk of Major Adverse Cardiovascular and Limb Events



Restricted cubic spline analyses show dose-response relationships between postdiagnosis smoking behavior and hazard ratios for major adverse cardiovascular and cerebrovascular events (MACCEs) and major adverse limb events (MALEs). A 0% reduction (maintained smoking) (A and B) and 0 cigarettes per day (complete cessation) (C and D) are the reference values. Shaded areas represent 95% CIs; dashed orange line, hazard ratio of 1.0.

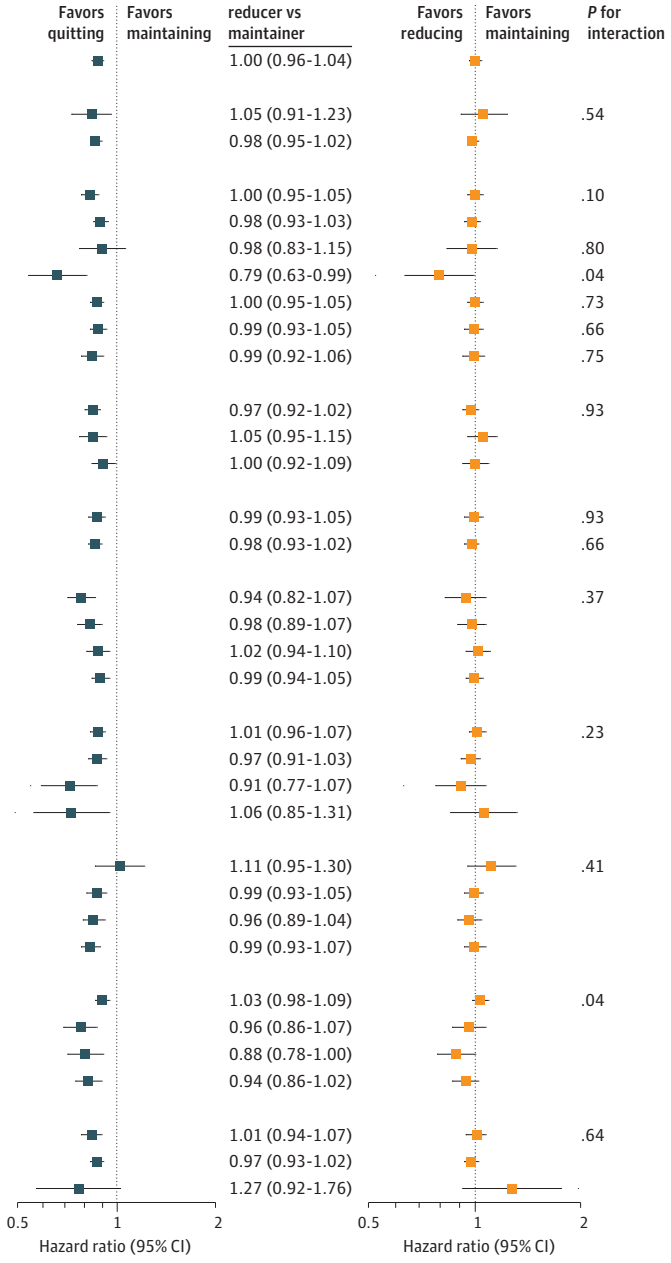
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cessation was not associated with lower risk of either outcome. These findings were robust to sensitivity analyses, including doubly robust estimation and multiple imputation, and remained consistent across demographic and clinical subgroups.

This study provides PAD-specific evidence on the association between postdiagnosis smoking cessation and cardiovascular and limb outcomes. Although smoking cessation benefits after coronary artery disease are well established,[28-32] their relevance in PAD requires direct evaluation. PAD and coronary artery disease share an underlying atherosclerotic process but differ in their

Figure 3. Forest Plot of the Association Between Postdiagnosis Smoking Behavior Change and Major Adverse Cardiovascular and Cerebrovascular Events (MACCEs)

Group	No. of events/ total No. (%) of events	MACCE IPT-weighted hazard ratio (95% CI), quitter vs maintainer	MACCE IPT-weighted hazard ratio (95% CI), reducer vs maintainer	*P* for interaction
Overall	18 502/188 125 (9.8)	0.88 (0.84-0.91)	1.00 (0.96-1.04)	
Sex				
Female	1246/21 736 (5.7)	0.84 (0.73-0.96)	1.05 (0.91-1.23)	.54
Male	17 256/166 388 (10.4)	0.86 (0.83-0.90)	0.98 (0.95-1.02)	
Age, y				
<65	9387/151 491 (6.2)	0.83 (0.78-0.88)	1.00 (0.95-1.05)	.10
≥65	9114/36 634 (24.9)	0.89 (0.85-0.94)	0.98 (0.93-1.03)	
MACCE history	1052/3669 (28.7)	0.90 (0.77-1.06)	0.98 (0.83-1.15)	.80
Peripheral revascularization history	590/1955 (30.2)	0.66 (0.54-0.81)	0.79 (0.63-0.99)	.04
Hypertension	11 761/83 836 (14.0)	0.87 (0.83-0.91)	1.00 (0.95-1.05)	.73
Diabetes without complications	8202/57 557 (14.3)	0.88 (0.83-0.93)	0.99 (0.93-1.05)	.66
Diabetes with complications	5402/32 400 (16.7)	0.84 (0.78-0.91)	0.99 (0.92-1.06)	.75
CCI				
0 or 1	11 942/142 509 (8.4)	0.85 (0.80-0.89)	0.97 (0.92-1.02)	.93
2	2983/25 894 (11.5)	0.85 (0.77-0.93)	1.05 (0.95-1.15)	
≥3	3577/19 721 (18.1)	0.91 (0.84-0.99)	1.00 (0.92-1.09)	
Medications				
Statin	7200/62 428 (11.5)	0.87 (0.82-0.92)	0.99 (0.93-1.05)	.93
Antithrombotic	13 392/116 394 (11.5)	0.86 (0.82-0.90)	0.98 (0.93-1.02)	.66
Smoking history, person-years				
0-10	2524/45 157 (5.6)	0.78 (0.71-0.86)	0.94 (0.82-1.07)	.37
10-20	3732/49 144 (7.6)	0.83 (0.76-0.90)	0.98 (0.89-1.07)	
20-30	3937/38 395 (10.3)	0.88 (0.81-0.95)	1.02 (0.94-1.10)	
≥30	8307/55 429 (15.0)	0.89 (0.84-0.95)	0.99 (0.94-1.05)	
Alcohol				
Nondrinker	9180/64 001 (14.3)	0.88 (0.83-0.92)	1.01 (0.96-1.07)	.23
Low risk	7553/101 963 (7.4)	0.87 (0.82-0.93)	0.97 (0.91-1.03)	
Moderate risk	1106/14 002 (7.9)	0.72 (0.59-0.87)	0.91 (0.77-1.07)	
Heavy risk	662/8159 (8.1)	0.73 (0.56-0.95)	1.06 (0.85-1.31)	
BMI				
<18.5	1029/5024 (20.5)	1.02 (0.86-1.21)	1.11 (0.95-1.30)	.41
18.5-22.9	7205/58 925 (12.2)	0.87 (0.81-0.93)	0.99 (0.93-1.05)	
23-24.9	4503/47 091 (9.6)	0.85 (0.79-0.92)	0.96 (0.89-1.04)	
≥25	5764/77 086 (7.5)	0.83 (0.78-0.89)	0.99 (0.93-1.07)	
Physical activity, d				
None	10 609/85 535 (12.4)	0.90 (0.86-0.95)	1.03 (0.98-1.09)	.04
1-2	2345/32 831 (7.1)	0.78 (0.69-0.87)	0.96 (0.86-1.07)	
3-4	1971/27 085 (7.3)	0.80 (0.71-0.91)	0.88 (0.78-1.00)	
≥5	3576/42 674 (8.4)	0.82 (0.75-0.90)	0.94 (0.86-1.02)	
eGFR, mL/min				
≥90	6294/86 179 (7.3)	0.84 (0.78-0.90)	1.01 (0.94-1.07)	.64
30-89	11 903/101 258 (11.8)	0.87 (0.83-0.91)	0.97 (0.93-1.02)	
<30	305/688 (44.3)	0.77 (0.57-1.03)	1.27 (0.92-1.76)	



This forest plot shows hazard ratios and 95% CIs for the association between postdiagnosis smoking behavior change and MACCE risk across predefined subgroups. Hazard ratios are from inverse probability of treatment (IPT)–weighted Cox proportional hazards regression models. BMI indicates body mass index (calculated as weight in kilograms divided by height in meters squared); CCI, Charlson comorbidity index; eGFR, estimated glomerular filtration rate.

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pathogenic mechanisms,[33] and smoking shows a stronger dose-response relationship with PAD incidence than with other atherosclerotic diseases.[13] These distinctions underscore the importance of this PAD-specific evidence for smoking cessation. Prior studies in patients with PAD have focused mainly on those undergoing revascularization. Reitz et al[12] demonstrated that preprocedural smoking status affected 30-day mortality and complications in 14 350 Veterans Affairs patients with PAD, whereas Armstrong et al[20] showed improved 5-year mortality and amputation-free survival among 204 active smokers who quit within 1 year after the procedure. Our findings of reduced MACCE risk in patients undergoing revascularization align with these observations, although we did not observe a significant reduction in MALE risk, likely reflecting the limited size of the revascularized subgroup or irreversible peripheral vascular damage. Importantly, our study extends existing evidence beyond patients undergoing revascularization, encompassing the broader spectrum of PAD severity, from asymptomatic disease to limb-threatening ischemia.

Notably, partial smoking reduction was not associated with lower risk of either MACCEs or MALEs. This finding aligns with evidence from the general population[34] and patients with coronary disease[31] and extends this observation to limb-specific outcomes in patients with PAD. This is biologically plausible given that even minimal smoking exposure (as low as 1 cigarette daily) is associated with substantially increased cardiovascular risk.[35] Our dose-response analyses showed that risk reduction was evident only for near-complete cessation; across most of the reduction range, continued smoking at any level remained associated with elevated risk. Although the spline curves suggested a gradual decrease beginning at approximately 75% reduction, estimates in this range are based on a small number of patients and are influenced by the smoothing properties of the spline model toward complete cessation. In the peripheral arterial bed, smoking promotes disease progression through endothelial dysfunction, oxidative stress, inflammation, arterial stiffness, and prothrombotic activation.[36] Although complete cessation may partially reverse these processes, as shown by measurable improvements in inflammatory and endothelial biomarkers within weeks to months of quitting,[37] the recovery is incomplete.[38] Considering the nonlinear cardiovascular toxic effects,[35,39,40] continued smoking even at a low level can be detrimental to this partial recovery, which may explain why only complete cessation was associated with lower event rates in our cohort. These findings have direct clinical implications: partial reduction should not be accepted as sufficient, and clinicians must emphasize complete cessation as the primary therapeutic goal.

The association between cessation and lower MACCE risk was consistent across most subgroups, including patients 65 years or older, those with diabetes, or those with prior myocardial infarction (MACCE incidence, 15%-25%). Two significant effect modifications were identified: cessation showed a greater relative risk reduction among physically active patients than among those with a sedentary lifestyle (*P* for interaction = 0.04), consistent with evidence of an additive interaction between smoking and physical inactivity on cardiovascular mortality,[41] and patients with prior peripheral revascularization showed a particularly strong association with cessation (*P* for interaction = 0.04). For MALEs, quitting was associated with a consistent pattern of lower risk across most subgroups (HR range, 0.51-0.91) (eFigure 5 in Supplement 1). However, the association was not significant among patients with prior MACCEs, prior peripheral revascularization, or higher comorbidity burden, possibly reflecting irreversible vascular damage in these subgroups, although limited event numbers preclude definitive conclusions. We also observed a gradient in the association between cessation and MACCEs by baseline pack-year burden, with greater risk reduction among those with lower cumulative exposure, although this did not reach statistical significance for interaction. Cessation was nonetheless associated with lower risk even among patients with a smoking history of 30 pack-years or more, supporting cessation efforts regardless of cumulative smoking history.[42,43]

Our findings also carry important clinical implications regarding smoking behavior after PAD diagnosis. Despite demonstrated benefits and guideline recommendations,[14-17] only 20% of active smokers in our cohort achieved complete cessation after PAD diagnosis. This rate is lower than that reported in studies of symptomatic patients or patients with PAD undergoing revascularization,[20,42]

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potentially reflecting lower perceived disease severity among those with earlier-stage disease or insufficient cessation support in clinical practice.[42-44] Current evidence for PAD-specific cessation interventions remains limited,[45] although previous work[46] in other populations has shown that multimodal approaches combining pharmacotherapy, behavioral support, and structured follow-up achieve higher sustained cessation rates. The association between cessation and lower event rates observed in our cohort provides further justification for developing intensive, tailored cessation programs for patients with PAD.

Limitations

This study has certain limitations that should be acknowledged. First, although our case definition was adapted from prior NHIS-based PAD research and combined diagnostic codes with concurrent antithrombotic prescriptions to enhance specificity, no formal validation study has assessed the diagnostic accuracy of PAD-related *ICD-10* codes in the Korean NHIS-NHID. Some degree of misclassification, both inclusion of patients without true atherosclerotic PAD and missed cases not captured by our criteria, cannot be excluded. Moreover, the requirement for health screening participation both before and after PAD diagnosis may have introduced healthy-participant bias, selecting for patients with less advanced disease. This likely contributed to the low rate of major amputation observed (n = 58), although this figure is consistent with prior Korean NHIS-based PAD studies[25,26] reporting similarly low rates. Future studies incorporating broader diagnostic criteria and populations not selected by health screening participation would help confirm whether these findings extend to patients with more advanced disease. Second, smoking status was self-reported without biochemical verification, which may introduce reporting bias. However, we assessed changes in smoking behavior rather than absolute consumption levels, and previous studies[47-49] suggest a reasonable correlation between self-reported smoking status and objective measures. Third, we lacked information on electronic tobacco product use and environmental smoke exposure, which may have influenced outcomes. Fourth, despite doubly robust adjustment, unmeasured confounding from concurrent lifestyle modifications or treatments cannot be ruled out. Fifth, some outcomes and subgroup analyses had limited statistical power due to low event rates, with several MALE subgroup analyses showing wide CIs. Additionally, the intervals between examinations (typically 2 years) may not capture short-term fluctuations in smoking behavior.

Conclusions

In this cohort study of 189 545 active smokers with newly diagnosed PAD, complete smoking cessation was associated with a lower risk of both MACCEs and MALEs, whereas partial smoking reduction was not associated with significant clinical benefit. These findings underscore the importance of complete cessation as the primary therapeutic goal in PAD management, and the low cessation rate observed in this cohort highlights the need for more effective, PAD-specific cessation interventions.

ARTICLE INFORMATION

Accepted for Publication: June 18, 2026.

Published: August 19, 2026. doi:10.1001/jamanetworkopen.2026.29337

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Conflict of Interest Disclosures: Dr Ha reported receiving grants and personal fees from Chong Kun Dang Pharmaceutical Corporation, Seoul, Korea and personal fees from Seoul National University Hospital outside the submitted work. No other disclosure were reported.

Funding/Support: This research was supported by grant 0420230730 from the Seoul National University Hospital Research Fund (Dr Han).

Role of the Funder/Sponsor: The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

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SUPPLEMENT 1.
eMethods 1. Data source
eMethods 2. Statistical analysis
eFigure 1. Study design
eFigure 2. Cohort selection flowchart
eFigure 3. Association between post-diagnosis smoking behavior change and individual components of major adverse cardiovascular and limb events
eFigure 4. Cumulative incidence of individual component outcomes of MACCE and MALE by post-diagnosis smoking behavior change in patients with incident peripheral artery disease
eFigure 5. Subgroup analysis on the association between post-diagnosis smoking behavior change and major adverse limb event
eTable 1. ICD-10/ HIRA EDI codes for cohort and outcome definition
eTable 2. ICD-10 codes for operational definition of comorbidities
eTable 3. Pre-diagnosis characteristics by post-diagnosis smoking behavior change: complete covariate list (unweighted)
eTable 4. Pre-diagnosis characteristics by post-diagnosis smoking behavior change: complete covariate list (SMD after IPTW)
eTable 5. Time intervals between health examinations and peripheral artery disease diagnosis by post-diagnosis smoking behavior group
eTable 6. Sensitivity analyses in the entire cohort using alternative missing-data methods
eTable 7. Sensitivity analyses using alternative smoking reduction thresholds (≥25% and ≥50%) for the reducer category
eTable 8. Sensitivity analysis excluding patients with prior cardiovascular or cerebrovascular events

SUPPLEMENT 2.
Data Sharing Statement

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